NIKITA KALININ

Visit my homepage | nikita.kalinin@ist.ac.at | Klosterneuburg Austria

EDUCATION

Institute of Science and Technology Austria (ISTA) — Klosterneuburg, Austria
PhD student in Machine Learning, advised by Prof. Christoph Lampert — Sep 2022 – Aug 2027

Higher School of Economics (HSE) — Moscow, Russia
BS in Computer Science GPA 9.12/10, top 2% — Sep 2018 – May 2022

Yandex School of Data Analysis (YSDA) — Moscow, Russia
Master's-level program in Data Science GPA 4.93/5, top 4% — Sep 2019 – May 2021

University of Helsinki — Helsinki, Finland
Exchange student in the Master's program in Computer Science — Jan 2021 – Jun 2021

SELECTED PUBLICATIONS

Google Scholar statistics: 18 papers, 83 citations, h-index of 5

1. Pavel Arkhipov, **Nikita P. Kalinin** "Improved Error Bounds for Pure Differentially Private Continual Counting via Matrix Factorization" ACM-SIAM Symposium on Discrete Algorithms **(SODA 2027)**
2. Jan Schuchardt, **Nikita P. Kalinin** "Sampling-free privacy accounting for matrix mechanisms under random allocation" Conference on Neural Information Processing Systems **(NeurIPS 2026)**
3. **Nikita P. Kalinin**, Ryan McKenna, Jalaj Upadhyay, and Christoph H. Lampert "Back to Square Roots: An Optimal Bound on the Matrix Factorization Error for Multi-Epoch Differentially Private SGD" International Conference on Learning Representations (**ICLR 2026**)
4. **Nikita P. Kalinin**, Joel Daniel Andersson "Learning Rate Scheduling with Matrix Factorization for Private Training" Foundations of Responsible Computing (**FORC 2026**)
5. Monika Henzinger, **Nikita P. Kalinin**, Jalaj Upadhyay "Normalized Square Root: Sharper Matrix Factorization Bounds for Differentially Private Continual Counting" Foundations of Responsible Computing (**FORC 2026**)
6. **Nikita P. Kalinin**, Jalaj Upadhyay, and Christoph H. Lampert. "Continual Release Moment Estimation with Differential Privacy". Conference on Neural Information Processing Systems (**NeurIPS 2025**)
7. **Nikita P. Kalinin** and Lukas Steinberger "Efficient Estimation of a Gaussian Mean with Local Differential Privacy" Conference on Artificial Intelligence and Statistics (**AISTATS 2025)**
8. **Nikita P. Kalinin** and Christoph H. Lampert "Banded Square Root Matrix Factorization for Differentially Private Model Training." Conference on Neural Information Processing Systems (**NeurIPS 2024)**

RESEARCH & INDUSTRY EXPERIENCE

University of Copenhagen — Copenhagen, Denmark
Visiting Student Researcher supervised by Prof. Rasmus Pagh — Aug 2025 – Oct 2025

- Collaborated on learning rate aware matrix factorization for differentially private training.

Huawei Moscow Research Center — Moscow, Russia
ML Engineer in Computer Vision Laboratory — Sep 2021 – Aug 2022

- Developed a reinforcement learning-based solution for autonomous vehicle parking.

Institute of Science and Technology Austria (ISTA) — Klosterneuburg, Austria
Scientific summer intern supervised by Prof. Marco Mondelli — Jun 2021 – Sep 2021

- Evaluated gradient compression algorithms against information theoretic lower bounds.

Laboratory of Methods for Big Data Analysis (LAMBDA) HSE — Moscow, Russia
Scientific intern working on active learning for GNN-based molecular potentials — Oct 2020 – May 2021

Yandex — Moscow, Russia
Machine learning developer intern in Media services — Jul 2020 – Oct 2020

Yandex — Moscow, Russia
Data analyst intern in push notification team — Jul 2019 – Nov 2019

INVITED TALKS

1. Imperial College London, UK: Efficient noise correlation for DP training. 6 Jul 2026.
2. Inria Montpellier, France: Efficient noise correlation for DP training. 26 May 2026.
3. Morgan Stanley (ML Research External Speaker): Practical noise correlation for DP training. 13 May 2026.
4. Inria Lille, France: Efficient noise correlation for DP training. 19 Feb 2026.
5. University of Helsinki, Finland: Efficient noise correlation for DP training. 9 Feb 2026.
6. University of Copenhagen, Denmark: Matrix factorization for multi-epoch DP-SGD. 29 Aug 2025.
7. CISPA Helmholtz Center, Germany: Matrix factorization for multi-epoch DP-SGD. 26 Jun 2025.
8. Google (Quirk Learning Team Meeting): Matrix factorization for multi-epoch DP-SGD. 5 Jun 2025.
9. Google (Privacy for ML Talk): Continual-release moment estimation. 16 Apr 2025.
10. TU Munich, Germany: Banded square-root factorization for DP training. 16 Jan 2025.
11. Google (Privacy for ML Talk): Banded square-root factorization for DP training. 12 Jun 2024.

ACADEMIC SERVICE

1. Conference Reviewer: NeurIPS (2025, 2026), ICLR (2027), ICML (2026 Gold Reviewer), ITCS (2027), SatML (2026, 2027), GCPR (2024, 2025)
2. Journal Reviewer: JASA (2025), TMLR (2026)
3. Mentorship:
   - Mentored ISTA intern Ali Najar on the project "Continual Mean Estimation under User-Level Differential Privacy" (Jul 2025 – Oct 2025).
   - Co-mentored ISTA intern Mihaela Hudisteanu (with Dr. Edwidge Cyffers) on "DP-MicroAdam: Private and Frugal Algorithm for Training and Fine-tuning" (Jul–Oct 2025). (**WiML Workshop at NeurIPS 2025**)
4. Workshop Organization: Co-organized the "Privacy-Preserving Machine Learning: Towards Tight Guarantees" workshop at **EurIPS 2025**. Coordinated speaker invitations, program organization, reviewer assignments, and communications.
5. Open Source: Contributed code to `jax_privacy`, Google DeepMind's open-source DP ML library.

TEACHING EXPERIENCE

1. ISTA (2025) — Guest lecture in the graduate-level course *Trustworthy Machine Learning* on amplification by subsampling and correlated noise for DP-SGD.
2. University of Copenhagen (2025) — Guest lecture in the graduate-level course *Advanced Topics in Machine Learning (ATML)* on matrix factorization for private machine learning.
3. ISTA — Teaching assistant for *Monte Carlo Methods in Statistical Physics* (2023), *Modern Machine Learning* (2024), and *Algorithms with Differential Privacy* (2025).
4. GoTo Summer School (2022) — Taught a week-long course on Reinforcement Learning to high school students.
5. Yandex School of Data Analysis (2020–2023) — Led recitation sessions and graded assignments for the *Machine Learning* course, working with undergraduate and master's students.
6. Seasonal Mathematics School Mediana (2019–2021) — Delivered lectures on Olympiad mathematics to advanced high school students.

TECHNICAL SKILLS

Programming: Python, C++, SQL
ML Frameworks: PyTorch, JAX, scikit-learn

LANGUAGES

Russian (native), English (C1, IELTS 7.5), German (B2)

HONORS AND AWARDS

First Prize, International Mathematics Competition for University Students (IMC), 2020
Third Prize, ACM ICPC Moscow Regional Contest, 2020
Segalovich Scholarship for Academic Excellence, 2019 and 2020
Silver Medal, Putnam Mirror Competition, 2019
Silver Medal, Romanian Master of Physics, 2017 (Russian national team)
Prize winner, National Physics Olympiad, 2016, 2017, 2018